From Geometric Flux to Magnetic Charge

A Checked Pauli Curvature and an Explicit Physical Bridge

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Abstract

A parameter-space monopole becomes a magnetic-charge statement only through a map to physical space and an identification of the measured field. We make that distinction concrete for the lower eigenprojector of a two-level Pauli Hamiltonian. A Lean module proves the Pauli commutator, the projector's exact affine increment, six direction-component derivatives, a general entrywise chain rule, both projector derivative specializations, and the curvature built from those actual derivatives, giving $\Omega_{\theta\phi} = \sin\theta/2$. Earlier checked modules integrate this coefficient to unit normalized flux, verify the two local coefficient identities, and prove the axis-extension obstruction. We then state the physical bridge $F_B = \kappa f^*\Omega$, which gives $\Phi_B = 2\pi\kappa \deg f$ under the usual degree formula. Simple numerical records expose precisely which implication depends on that bridge. The resulting argument connects the explicit matrix calculation to its flux and keeps the physical coupling visible. Bundle construction, Chern-class identification and the physical pullback remain ordinary written mathematics.

Keywords: Berry curvature; Pauli projector; magnetic monopole; first Chern number; physical identification; Lean 4.

1 From a projector to a physical question

Start with an eigenline that changes as the Hamiltonian changes. Its curvature records the geometry of that family. Integrating the curvature can retain an integer even though no magnetic field in physical space has yet been specified. The question is what carries that integer from one domain to the other.

The two-level Pauli example lets us follow each step explicitly: matrix, derivative, curvature, flux, and physical interpretation. The new formal step in this revision is the matrix-to-curvature calculation. Its sign and factor of one half follow from checked algebra and trigonometric derivatives. The calculation supplies the coefficient integrated by the earlier development.

The underlying geometry is classical. Dirac's construction concerns electromagnetic phase and magnetic charge [1]; Berry and Simon describe eigenstate phase and line-bundle geometry [2, 3]; Wu and Yang explain the global connection picture [4]. The example also appears in the author's IQGT manuscript, Appendix B [5]. This note's contribution is a reproducible formal calculation and a clear account of the physical identification it needs.

We mark checked components and written proofs separately. The declaration map in Appendix A gives the exact formal coverage. All physical interpretations use the domains, conventions and hypotheses stated below.

2 Conventions and domains

Let $\sigma_1, \sigma_2, \sigma_3$ be the standard Pauli matrices, satisfying

$$(a \cdot \sigma)(b \cdot \sigma) = (a \cdot b)I + \mathrm{i}(a \times b) \cdot \sigma, \qquad \operatorname{Tr} \sigma_j = 0. \tag{1}$$

For $d \in \mathbb{R}^3 \setminus \{0\}$ write $n = d/|d|$ and consider

$$H(d) = d \cdot \sigma, \qquad P_-(n) = \tfrac{1}{2}(I - n \cdot \sigma). \tag{2}$$

The base space at this stage is Hamiltonian *parameter space*. On its unit sphere use

$$n(\theta, \phi) = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta), \quad 0 \le \theta \le \pi, \quad 0 \le \phi < 2\pi. \tag{3}$$

The outward orientation is represented by $\mathrm{d}\theta \wedge \mathrm{d}\phi$ with its positive $\sin\theta$ surface density. For a normalized local eigenvector u, our Berry convention is

$$A = \mathrm{i}u^\dagger \mathrm{d}u, \qquad \Omega = \mathrm{d}A, \qquad \Omega_{ab} = \mathrm{i}\operatorname{Tr}(P_-[\partial_a P_-, \partial_b P_-]). \tag{4}$$

Changing the sign convention for A, the band, or the orientation changes the corresponding sign. The nonvanishing statement is unaffected.

The magnetic two-form F_B used later is a dimensionful physical quantity whose surface integral is magnetic flux. It is not defined to be Ω. A coefficient converting between them must carry the requisite units, and a map must identify the parameter and spatial domains. We use q_Φ for a nonzero *flux unit*; it is not an electric charge.

3 Projector algebra and derived curvature

Proposition 3.1 (Pauli projector identities)**.** [Lean-checked components] *For real x, y, z with $x^2 + y^2 + z^2 = 1$, the matrix*

$$P(x, y, z) = \frac{1}{2}\begin{pmatrix} 1 - z & -x + \mathrm{i}y \\ -x - \mathrm{i}y & 1 + z \end{pmatrix} \tag{5}$$

is Hermitian and idempotent and has trace one.

Proof. Conjugation exchanges the off-diagonal entries. Put $N = n \cdot \sigma$. Equation (1) gives $N^2 = I$, so $P^2 = (I - 2N + N^2)/4 = P$, and $\operatorname{Tr} P = 1$. The companion development proves these three matrix identities directly in fixed coordinates over $\mathbb{C}$; it does not rely on an assumed matrix identity named (1). □

In ordinary finite-dimensional spectral theory these identities specify an orthogonal rank-one projector. Moreover, $HP_- = -|d|P_-$ follows directly from (2). The rank conclusion and this eigenvalue relation are written consequences here, not additional checked declarations.

Proposition 3.2 (Berry curvature of the lower eigenline)**.** [Lean-checked components] *With the convention* (4)*,*

$$\Omega_{ab} = \tfrac{1}{2} n \cdot (\partial_a n \times \partial_b n), \qquad \Omega = \tfrac{1}{2} \sin\theta \, \mathrm{d}\theta \wedge \mathrm{d}\phi. \tag{6}$$

Proof. Set $a = \partial_a n$, $b = \partial_b n$, and $c = a \times b$. Differentiating $P_- = (I - n \cdot \sigma)/2$ and using (1) gives

$$[\partial_a P_-, \partial_b P_-] = \frac{1}{4}[a \cdot \sigma, b \cdot \sigma] = \frac{\mathrm{i}}{2} c \cdot \sigma.$$

Since $\mathrm{Tr}((n \cdot \sigma)(c \cdot \sigma)) = 2n \cdot c$,

$$\mathrm{Tr}(P_-[\partial_a P_-, \partial_b P_-]) = -\frac{\mathrm{i}}{2} n \cdot c.$$

Multiplication by i proves the first formula. For (3), direct differentiation gives $\partial_\theta n \times \partial_\phi n = \sin\theta \, n$, yielding the second. □

The new module `IqgtGrassmannian/Monopole.lean` checks this calculation in fixed 2×2 complex matrices. It proves the commutator for arbitrary real vectors, the exact increment

$$P(n + ha) - P(n) = hD(a), \qquad D(a) = -\tfrac{1}{2} a \cdot \sigma,$$

and the trace identity $\mathrm{i}\,\mathrm{Tr}(P(n)[D(a), D(b)]) = n \cdot (a \times b)/2$. Six `HasDerivAt` theorems identify the components of $\partial_\theta n$ and $\partial_\phi n$. The general lemma `hasDerivAt_proj_comp` differentiates every entry of the affine Pauli matrix formula along an arbitrary real path with differentiable components. Its two specializations `hasDerivAt_proj_theta` and `hasDerivAt_proj_phi` then prove, for every pair of matrix indices,

$$\frac{\mathrm{d}}{\mathrm{d}\theta} P(n(\theta,\phi))_{ij} = D(\partial_\theta n)_{ij}, \qquad \frac{\mathrm{d}}{\mathrm{d}\phi} P(n(\theta,\phi))_{ij} = D(\partial_\phi n)_{ij}.$$

These are derivatives of complex-valued entry functions, requiring no choice of a matrix norm. Finally, `berry_curvature_from_derivatives` builds the two derivative matrices entry by entry using Lean's `deriv` operator and proves that their matrix-trace curvature equals $\omega(\theta) = \sin\theta/2$. The derivative link and final trace calculation are thus joined by a checked theorem. The table transliterates the Unicode direction characters in the derivative theorem names. An exterior-form construction remains outside the module.

The new and original projects use the same explicit projector entries and the same scalar formula $\omega(\theta) = \sin\theta/2$. They are verified in separately pinned environments. Their identification here is by the displayed formulas; this release does not claim a cross-project Lean import theorem.

Proposition 3.3 (Normalized flux)**.** [Lean-checked components] *Define the iterated coordinate flux and its normalization by*

$$\mathcal{F}(f) = \int_0^{2\pi} \mathrm{d}\phi \int_0^{\pi} f(\theta) \, \mathrm{d}\theta, \qquad \mathcal{N}(f) = \frac{\mathcal{F}(f)}{2\pi}. \tag{7}$$

For $\omega(\theta) = \sin\theta/2$*,*

$$\int_0^{\pi} \omega(\theta) \, \mathrm{d}\theta = 1, \quad \mathcal{F}(\omega) = 2\pi, \quad \mathcal{N}(\omega) = 1 \neq 0. \tag{8}$$

Proof. The first integral equals $[-\cos\theta/2]_0^{\pi} = 1$. Integration in ϕ gives 2π, and division by the nonzero number 2π gives one. Lean uses Mathlib's interval integral of sine and exact real arithmetic. □

There is no additional surface Jacobian to insert into (7): the coefficient in (6) already contains $\sin\theta$. Multiplying it by another $\sin\theta$ would compute a different integral.

4 Two charts and the meaning of the integer

The familiar local eigenvectors make the geometric interpretation explicit. On the sphere with the south pole removed, and with the north pole removed, respectively, take

$$u_N = \begin{pmatrix} -e^{-\mathrm{i}\phi}\sin(\theta/2) \\ \cos(\theta/2) \end{pmatrix}, \qquad u_S = \begin{pmatrix} -\sin(\theta/2) \\ e^{\mathrm{i}\phi}\cos(\theta/2) \end{pmatrix}. \tag{9}$$

They satisfy $H(n)u_{N,S} = -u_{N,S}$ and $u_S = e^{\mathrm{i}\phi}u_N$ on the overlap. Computing (4) gives

$$A_N = a_N(\theta)\,\mathrm{d}\phi, \quad a_N = \frac{1-\cos\theta}{2}; \qquad A_S = a_S(\theta)\,\mathrm{d}\phi, \quad a_S = -\frac{1+\cos\theta}{2}. \tag{10}$$

The apparent polar-coordinate singularity of $\mathrm{d}\phi$ is cancelled by its coefficient at the included pole. The opposite pole is excluded from the respective chart. These are local descriptions of one connection, rather than one globally defined scalar potential on the entire sphere.

Proposition 4.1 (Local coefficient identities)**.** [Lean-checked components] *For every real θ,*

$$a_N'(\theta) = a_S'(\theta) = \omega(\theta), \qquad a_N(\theta) - a_S(\theta) = 1. \tag{11}$$

Proof. Differentiate (10) and subtract the two coefficients. These scalar derivative and equality statements are precisely the checked claims. □

The line bundle assembled from (9) has the usual first-Chern interpretation

$$c_1[P_-] = \frac{1}{2\pi}\int_{S^2}\Omega = 1 \tag{12}$$

with the specified convention. The transition phase on the equator has winding one. This is the standard eigenline-bundle interpretation, explained in writing using the local charts and classical connection theory [3, 4]. The companion code does not construct that bundle, its transition-function winding, or its Chern class. In particular, its name `normalizedFlux` denotes the real-valued expression in (7), not a formal Chern-class definition.

The complementary upper eigenline has $P_+ = I - P_-$. Its derivatives negate those of P_-, and the trace of a commutator is zero. Consequently

$$\mathrm{i}\,\mathrm{Tr}(P_+[\partial_a P_+, \partial_b P_+]) = -\Omega_{ab}.$$

Thus the full two-band bundle is trivial and its summed curvature vanishes, while either selected eigenline can have nonzero flux. Lean checks the integral cancellation $\mathcal{F}(\omega) + \mathcal{F}(-\omega) = 0$. The new module also checks the complement-projector trace identity, `complement_curvature`; the original integral theorem checks cancellation of the two supplied coefficients. Triviality of an ambient space cannot be substituted for triviality of each subbundle.

5 An extension obstruction at the degeneracy

At $d = 0$, $H(d)$ has no isolated lower eigenline. The obstruction can be seen without any Chern-class machinery. Restricting the normalized direction to the z-axis gives

$$P_-(0,0,1) = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \qquad P_-(0,0,-1) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}. \tag{13}$$

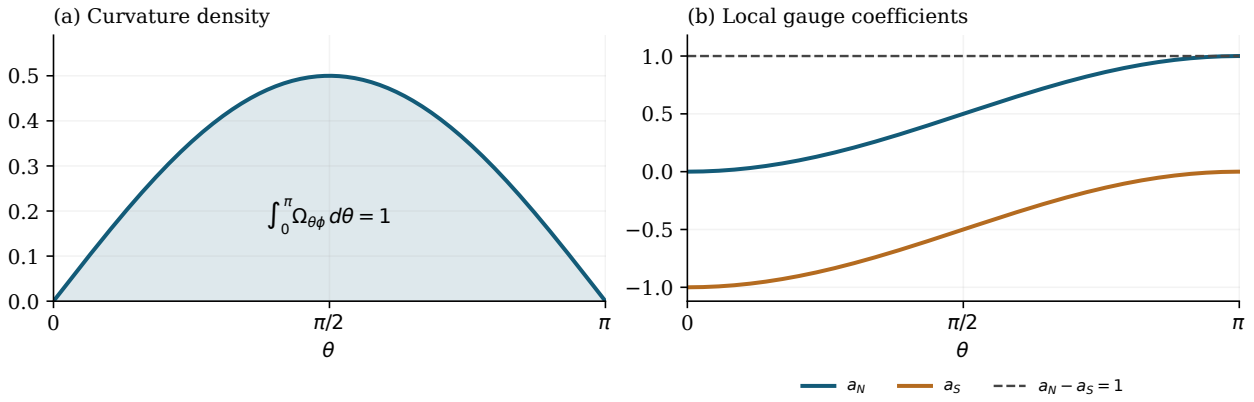


Figure 1: Exact analytic functions in the two-level example. (a) The polar integral of the displayed curvature coefficient is one. (b) The two local connection coefficients have the same derivative and differ by one. The plots illustrate formulas, not measured data; their scalar identities are checked in Lean.

Theorem 5.1 (No continuous axis extension)**.** [Lean-checked components] *Suppose* $F : \mathbb{R} \to M_2(\mathbb{C})$ *satisfies* $F(t) = P_-(0, 0, 1)$ *for* $t > 0$ *and* $F(t) = P_-(0, 0, -1)$ *for* $t < 0$*. Then* F *is not continuous at zero.*

Proof. The real part of its $(0, 0)$ entry is zero on the positive half-axis and one on the negative half-axis. Along $t_m = 1/(m+1)$ it tends to zero, and along $-t_m$ it tends to one. Both input sequences tend to zero. Continuity and uniqueness of limits would force $F(0)_{00}$ to have real part both zero and one, a contradiction. Lean proves the scalar jump obstruction and then applies it to the matrix coordinate. □

Any continuous three-dimensional extension of the normalized family that agrees with these axis values would restrict to such a continuous F; hence it cannot exist. That restriction argument and the three-dimensional spectral-family construction are stated in writing, while Theorem 5.1 is the exact checked result. This is a parameter-space degeneracy obstruction, not an electromagnetic source equation.

6 The electromagnetic bridge

6.1 A spatial domain and a map

Let Σ be a closed oriented physical spatial surface. A map $f : \Sigma \to S^2$ specifies how its points sample the Hamiltonian directions. It produces $f^*\Omega$, but this pullback is still an eigenstate-geometric two-form unless it is identified with the electromagnetic field measured by a stated probe.

A particularly transparent sufficient identification is

$$F_B|_\Sigma = \kappa f^*\Omega, \tag{14}$$

where κ is a nonzero dimensional coefficient. This equality is an additional hypothesis. A physical derivation must specify the coupling that makes the two curvatures the same observable, with consistent gauge transformations and normalization. Merely giving both quantities a connection or curvature notation does not supply that derivation.

Proposition 6.1 (Conditional pullback flux)**.** [Written proof; not formalized here] *If $\Sigma \simeq S^2$, f is smooth of degree d, and (14) holds, then*

$$\Phi_B := \int_\Sigma F_B = 2\pi\kappa d. \tag{15}$$

More generally, put $n = (2\pi)^{-1}\int_\Sigma f^*\Omega$ whenever this record is justified. Then $\Phi_B = q_\Phi n$ with $q_\Phi = 2\pi\kappa$.

Proof. The integral of the pullback of an oriented top-degree form under a smooth sphere map multiplies the integral by its degree. Apply that standard degree formula to (8) and multiply by κ. The second assertion is substitution. Neither the degree formula nor the pullback construction is formalized in the archive. □

A constant map has $d = 0$, even though the original parameter-sphere eigenline has $c_1 = 1$. An orientation-preserving degree-one map retains the unit normalized flux. Specifying a domain map is therefore essential even before specifying the electromagnetic coupling.

For a probe of nonzero signed electric charge e, a conventionally compatible physical identification of its dimensionless connection can set $\kappa = \hbar/e$. Under that *additional identification*, (15) reads $\Phi_B = (h/e)d$. This illustrates the usual Dirac flux unit; it neither derives the identification from the two-level projector nor reports an observed monopole. We keep flux separate from magnetic-source conventions that introduce factors such as μ_0.

6.2 An explicit exterior field

As a written example on punctured Euclidean space, choose $f(x) = x/|x|$ and impose (14). The associated radial magnetic field is

$$B(x) = \frac{\kappa}{2}\frac{\widehat{x}}{|x|^2}, \qquad x \neq 0. \tag{16}$$

Its flux is $2\pi\kappa$ through every centered sphere. This is the standard monopole exterior geometry, smooth away from the missing origin. It is a valid mathematical example conditional on the imposed field identification; it does not provide a nonsingular core, a source dynamics, or an empirical realization.

The exact formalized calculation is narrower. For the supplied unit-sphere density $g\sin\theta$, Lean proves

$$\mathcal{F}(g\sin\theta) = 4\pi g, \qquad \mathcal{F}(g\sin\theta) \neq 0 \iff g \neq 0. \tag{17}$$

Taking $g = \kappa/2$ reproduces the flux of (16). The code does not construct a spatial vector field or prove its Maxwell equations.

7 Charge transfer with an explicit bridge

A small numerical model isolates the role of the bridge.

Definition 7.1 (Charge record)**.** A record is a pair $r = (n, \Phi) \in \mathbb{Z} \times \mathbb{R}$. Write

$$G(r) \iff n \neq 0, \qquad M(r) \iff \Phi \neq 0.$$

For a specified flux unit q_Φ, define the bridge predicate

$$E(q_\Phi, r) \iff \Phi = q_\Phi n. \tag{18}$$

Here n is supplied as an integer; the type itself proves no integrality theorem. The code uses "magnetically charged" to mean $\Phi \neq 0$. The physical meaning of Φ comes from the preceding spatial and field identification.

Theorem 7.2 (Counterassignments without a bridge)**.** [Lean-checked components] *For every integer n, the records $r_0 = (n, 0)$ and $r_1 = (n, 1)$ have the same geometric coordinate, while $\neg M(r_0)$ and $M(r_1)$. In particular,*

$$\neg(\forall r,\ G(r) \Rightarrow M(r)), \tag{19}$$

$$\neg(\forall r,\ G(r) \Rightarrow \neg M(r)). \tag{20}$$

Proof. The first assertion is immediate from the coordinates. Taking $n = 1$ gives a counterexample to either universal implication. Lean supplies these explicit witnesses. □

The records $(1, 0)$ and $(1, 1)$ demonstrate that two freely assigned coordinates carry no relation until one is imposed. They are numerical witnesses, not physical solutions. Their purpose is to locate the missing premise.

Theorem 7.3 (Charge transfer under the bridge)**.** [Lean-checked components] *If $q_\Phi \neq 0$ and $E(q_\Phi, r)$, then*

$$M(r) \iff G(r). \tag{21}$$

Proof. Substitute $\Phi = q_\Phi n$. Since $\mathbb{R}$ has no zero divisors, $q_\Phi n \neq 0$ is equivalent to $n \neq 0$ when $q_\Phi \neq 0$. The embedding $\mathbb{Z} \hookrightarrow \mathbb{R}$ preserves zero. These are the steps checked by `bridge_transfers_charge`. □

The arithmetic is elementary. The physical work lies in deriving the bridge for the system and probe under study.

Two useful companion facts are checked. First, a unit geometric record with zero magnetic flux cannot satisfy a nonzero-unit bridge. Second, for any chosen q_Φ and n there exists the numerical bridge-compatible record $(n, q_\Phi n)$. These are consistency checks for the numerical definitions.

8 The limits of a no-flux argument

There is also a precise boundary to an absence claim. If a globally defined smooth one-form A exists on a closed oriented surface Σ and $F_B = \mathrm{d}A$ there, Stokes' theorem gives $\int_\Sigma F_B = 0$. Likewise, if a smooth closed field extends over a ball bounded by Σ, its boundary flux vanishes. These are written standard differential-form arguments, not theorems in this Lean package.

Their hypotheses matter. A nontrivial line bundle may require local potentials, and a punctured ball need not admit the extension required by the second argument. Local closedness $\mathrm{d}F_B = 0$ on the punctured exterior does not force all closed-surface fluxes there to vanish. The local charts in Section 4 demonstrate how a nonzero curvature integral can coexist with regular potentials on separate patches.

The code isolates the simplest logical version. Let $A_0(r)$ mean $\Phi = 0$. It proves

$$(\forall r,\ A_0(r) \Rightarrow \neg M(r)) \quad \text{and} \quad \neg(\forall r, \neg M(r)). \tag{22}$$

The first statement restricts an admitted class. The second uses a record outside that class. Thus imposing zero flux as an admissibility condition is different from deriving a universal no-monopole theorem. The numerical statement isolates the distinction between restricting a class and deriving a property of every member of a physical theory.

9 Relation to projector transport and retained information

The example illustrates a methodological point relevant to a transport atlas: identify the object, base space and observation before treating two invariants as the same quantity. Several distinctions are already visible in this note.

Record	What it retains here	What it does not supply by itself
Selected projector P_-	An eigenline at each parameter point	A physical spatial map or EM coupling
Scalar curvature integral	One oriented normalized flux	The full local field or dynamics
Summed two-band curvature	Cancellation of opposite contributions	Vanishing of each selected-band contribution
Geometric integer n	One topological/flux label under stated assumptions	A dimensional magnetic-flux unit
Candidate spatial flux Φ	An integrated physical reading once identified	A core solution, particle mass or production mechanism

A scalar summary can discard a distinction retained by a selected sector. Here the most immediate example is cancellation of two nonzero band fluxes in a trivial ambient bundle. That observation supports careful record selection; it does not identify Berry curvature, Hopf charge, instanton number or spectral flow with the same electromagnetic magnetic charge. Nor does the name "monopole" for an isotropic angular moment in a transport expansion perform such an identification.

For a light-transport, black-hole or neutrino model, the same discipline applies: specify the base space, retained sector, observable map and coupling before transferring an invariant. The present example supplies a worked template for that step.

10 Formal verification and reproducibility

The evidence comes from two separately pinned developments. The original archive contains 24 theorem declarations: 13 in `Geometry.lean`, two in `Extension.lean`, and nine in `Bridge.lean`. It uses Lean 4.19.0 and Mathlib commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b.`

Its build, axiom audit and six false controls passed. The controls target zero unit flux, either unconditional magnetic inference, omission of the nonzero flux unit, idempotence of $I/2$, and zero chart-coefficient difference. A separate kernel-replay tool was not run for this original archive.

The new `Monopole.lean` module contains 17 theorem declarations in the IQGT repository, branch `monopole-berry`, pinned to Lean 4.34.1. Its repository-wide check covers 111 declarations across five modules and eight false controls. It builds the modules, replays them using `leanchecker`, audits every named theorem's axiom dependencies and requires mathematical rejection of each false control. The two Pauli controls attempt to replace $\sin\theta/2$ with $\sin\theta$ and to reverse the sign of the top-left entry's θ derivative. The checked revision is commit `7851127`; the CI run and evidence identify the precise build. The module hash and resulting report accompany this revision.

Across both developments the allowed foundational dependencies are `propext`, `Classical.choice` and `Quot.sound`. The counts describe declarations; the 111 repository declarations include the new 17 and many IQGT results outside this note. They should not be added to the original 24 as though they were all claims of this paper.

The original archive rebuilds with `lake exe cache get` and `python3 scripts/verify.py`. To reproduce the new check, check out the exact IQGT commit recorded with this revision, fetch its pinned dependencies and run its verification workflow. The evidence and instructions distinguish the two environments and preserve the original archive unchanged.

11 What the checked calculation adds

The matrix-to-curvature step is now explicit. The lower Pauli matrix gives the positive coefficient $\sin\theta/2$ in the chosen convention; integrating that coefficient gives unit normalized flux. Selecting the complementary band reverses the curvature. This is the concrete geometric content that a physical application can carry forward.

The remaining mathematical work has distinct targets. A bundle formalization would connect the normalized integral to a first Chern class. A pullback formalization would carry the form to a specified spatial surface and track its degree. A physical field model would establish the coupling $F_B = \kappa f^*\Omega$ for a stated probe and domain. The checked finite-matrix calculation supports these tasks without supplying their hypotheses.

For the transport atlas, the practical rule is precise: keep the invariant with its domain and observation map. Berry flux, Hopf charge, instanton number and an $\ell = 0$ transport moment can each be useful while retaining different meanings. A successful bridge identifies the relevant structures and units in a particular system. That is where a geometric calculation becomes a testable magnetic-field claim.

A Complete declaration-to-claim map

All names below lie in the namespace `MonopoleStance`. The first table lists every declaration in the original archive. The second lists all 17 declarations in the new module. Local eigenvector chart construction, the Chern-class interpretation, physical pullbacks and Stokes arguments remain written mathematics.

Lean declaration	Exact role
Geometry: 13 declarations	
`lowerProjector_hermitian`	Hermiticity of the explicit matrix.
`lowerProjector_idempotent`	Idempotence under unit-norm hypothesis.
`lowerProjector_trace`	Trace equals one.
`omega_polar_integral`	Polar integral of supplied coefficient equals one.
`omega_sphereFlux`	Iterated flux equals 2π.
`omega_unit_normalizedFlux`	Normalized flux equals one.
`omega_nonzero_normalizedFlux`	Same normalized flux is nonzero.
`opposite_flux_cancels`	Supplied opposite coefficients have cancelling integrals.
`north_derivative`	Derivative of a_N equals ω.
`south_derivative`	Derivative of a_S equals ω.
`gauge_coefficient_difference`	$a_N - a_S = 1$.
`radialFlux_eq`	Supplied radial density integrates to $4\pi g$.
`radialFlux_nonzero_iff`	That flux is nonzero exactly when $g \neq 0$.

Lean declaration	Exact role
Extension: 2 declarations	
`no_continuous_jump_extension`	Distinct constant half-axis values preclude continuity.
`no_continuous_Pauli_axis_extension`	Matrix application using the two pole projectors.
Bridge: 9 declarations	
`bridge_transfers_charge`	Nonzero-unit bridge transfers nonzero charge.
`same_geometry_opposite_magnetic_status`	Same geometric coordinate; zero and unit flux.
`unit_geometry_compatible_with_both_records`	Both assignments at geometric charge one.
`no_unconditional_existence_inference`	Counterexample to universal existence inference in the record language.
`no_unconditional_absence_inference`	Counterexample to universal absence inference in the record language.
`unit_geometry_zero_flux_excludes_bridge`	Unit geometry and zero flux exclude a nonzero-unit bridge.
`bridge_record_exists`	Numerical record $(n, q_\Phi n)$ exists.
`admissibility_is_restriction`	Zero-flux restriction does not cover all records.
`formal_stance`	Conjunction of unit flux, both assignments and conditional transfer.

New matrix-curvature module: 17 declarations

Lean declaration	Exact role
`pauli_comm`	Pauli commutator for arbitrary real vectors.
`proj_affine`	Exact affine increment of the projector matrix.
`curvature_formula`	Matrix-trace curvature identity.
`hasDerivAt_nvec_theta1`	First component of $\partial_\theta n$.
`hasDerivAt_nvec_theta2`	Second component of $\partial_\theta n$.
`hasDerivAt_nvec_theta3`	Third component of $\partial_\theta n$.
`hasDerivAt_nvec_phi1`	First component of $\partial_\phi n$.
`hasDerivAt_nvec_phi2`	Second component of $\partial_\phi n$.
`hasDerivAt_nvec_phi3`	Third component of $\partial_\phi n$.
`hasDerivAt_proj_comp`	Entrywise chain rule along any real path with differentiable components.
`hasDerivAt_proj_theta`	Actual derivative of each projector entry in θ.
`hasDerivAt_proj_phi`	Actual derivative of each projector entry in ϕ.
`berry_curvature_from_derivatives`	Curvature built from actual entrywise derivatives.
`triple_product`	Spherical triple product equals $\sin\theta$.
`berry_curvature`	Trace coefficient equals $\omega(\theta)$.
`berry_curvature_eq_zero_iff`	$\omega(\theta) = 0$ iff $\sin\theta = 0$.
`complement_curvature`	Complementary matrix has opposite curvature.

The eight direction-labelled derivative names use the Unicode characters θ and ϕ in the Lean source; the table transliterates them for PDF font compatibility.

B Combined formal statement

The combined theorem, expressed mathematically, is

$$\begin{aligned}&\mathcal{N}(\omega) = 1\\ &\quad \wedge \left[(\exists r:\ n(r) = 1 \wedge \neg M(r)) \wedge (\exists r:\ n(r) = 1 \wedge M(r))\right]\\ &\quad \wedge \left[\forall q_\Phi \neq 0\ \forall r:\ E(q_\Phi, r) \Rightarrow (M(r) \Longleftrightarrow G(r))\right].\end{aligned}$$

The first conjunct concerns a real integral. The second concerns freely assigned records. No formal map from that integral to the integer field of a record is asserted by the conjunction. In the physical application that interpretation is an additional construction, discussed in Sections 4 and 6. The third conjunct is a transfer implication given a bridge. Reading all three as one unconditional physical-existence theorem would erase their different domains and hypotheses.

Data, code and assistance

No experimental dataset is introduced. Figure 1 plots analytic formulas. The companion source, locked build configuration, verification script and logs are supplied with this draft. The contributor names and AI disclosure appear above the abstract. The new source snapshot and exact CI evidence accompany the original proof archive.

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